

中国科学院大学
University of Chinese Academy of Sciences

Generative Models & AI4S Applications

Lecture 9 Generative Adversarial Networks I

Fall 2026

Liu Yunfan (刘云帆)

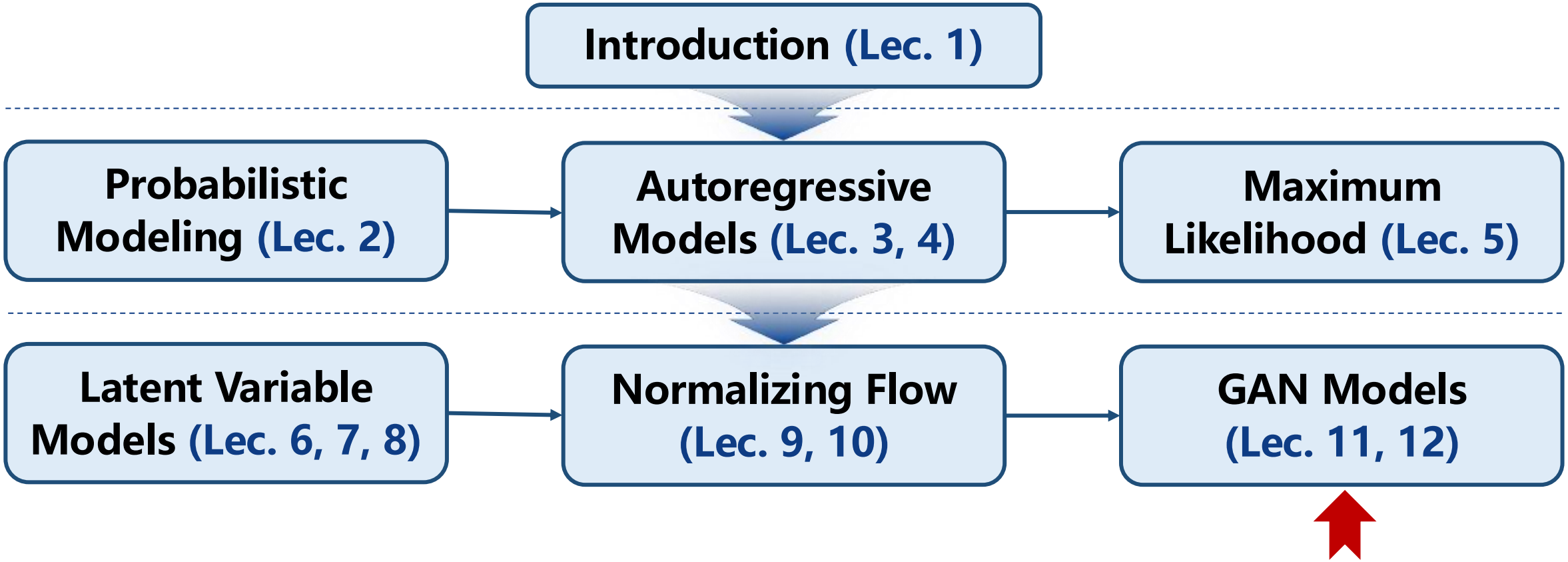
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Announcements

- Update on the description of Question 1 in Assignment 1

- One-week delay for Assignment 1 (due on Oct 25th)

Roadmap



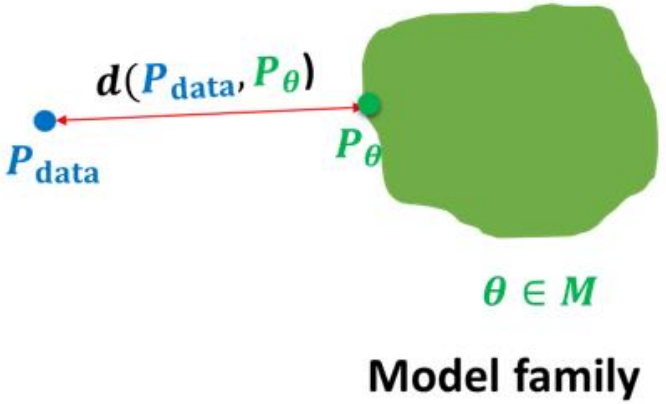
Introduction to various generative models

Recap

Images of dogs



$x_i \sim P_{\text{data}}$
$i = 1, 2, \ldots, n$



Dog Image Generator

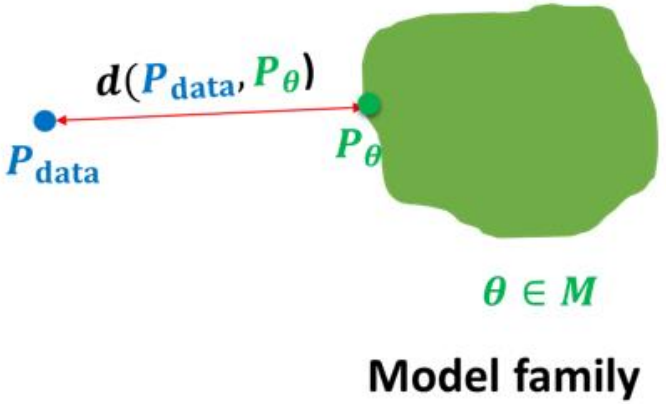
- Generative models learned so far:
  - **Autoregressive Models:** $p_\theta(x) = \prod_{i=1}^{n} p_\theta(x_i \mid x_{<i})$
  - **Variational Autoencoders:** $p_\theta(x) = \int p_\theta(x, z)\, dz$
  - **Normalizing Flow Models:** $p_X(x; \theta) = p_Z\left(f_\theta^{-1}(x)\right)\left|\det\left(\frac{\partial f_\theta^{-1}(x)}{\partial x}\right)\right|$
- Learning objective: **Maximum Likelihood / Minimize** $D_{KL}(p_\theta \,||\, p_{\text{data}})$

Recap

Images of dogs



$x_i \sim P_{\text{data}}$
$i = 1, 2, \ldots, n$



Dog Image Generator

- Generative models learned so far:
  - **Autoregressive Models:** $p_\theta(x) = \prod_{i=1}^{n} p_\theta(x_i \mid x_{<i})$
  - **Variational Autoencoders:** $p_\theta(x) = \int p_\theta(x, z)\, dz$
  - **Normalizing Flow Models:** $p_X(x; \theta) = p_Z\left(f_\theta^{-1}(x)\right)\left|\det\left(\frac{\partial f_\theta^{-1}(x)}{\partial x}\right)\right|$
- **However: Is it a good indicator of the quality of generation?**



Lecture Outline

- **Towards Likelihood-Free Training**
- **Generative Adversarial Networks (GANs)**
- **Property of GANs**

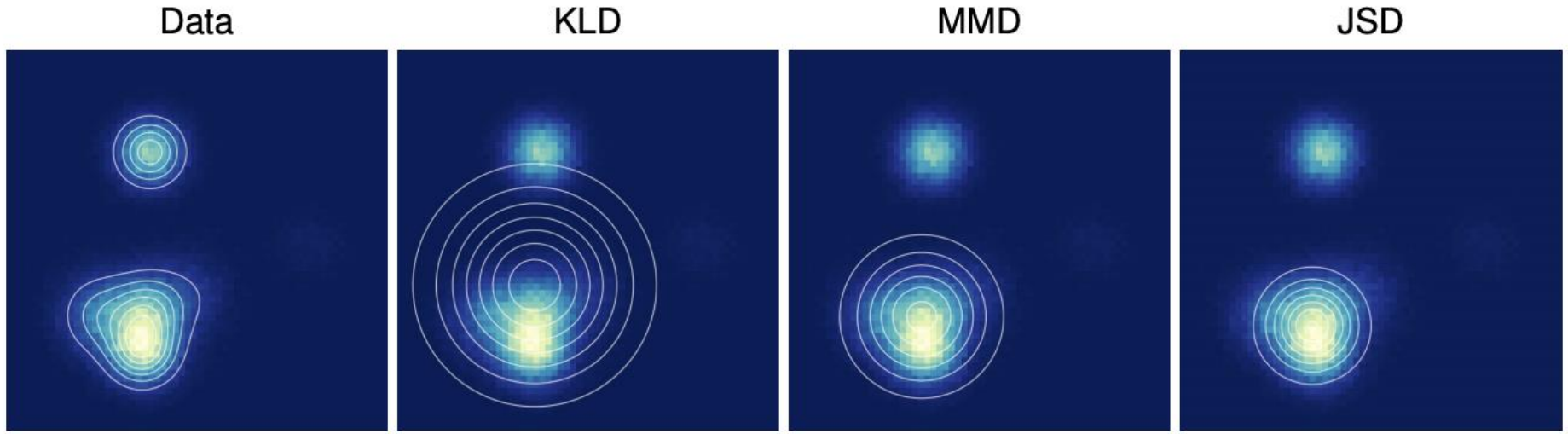


Lecture Outline

- **Towards Likelihood-Free Training**
- **Generative Adversarial Networks (GANs)**
- **Property of GANs**

Towards Likelihood-Free Training

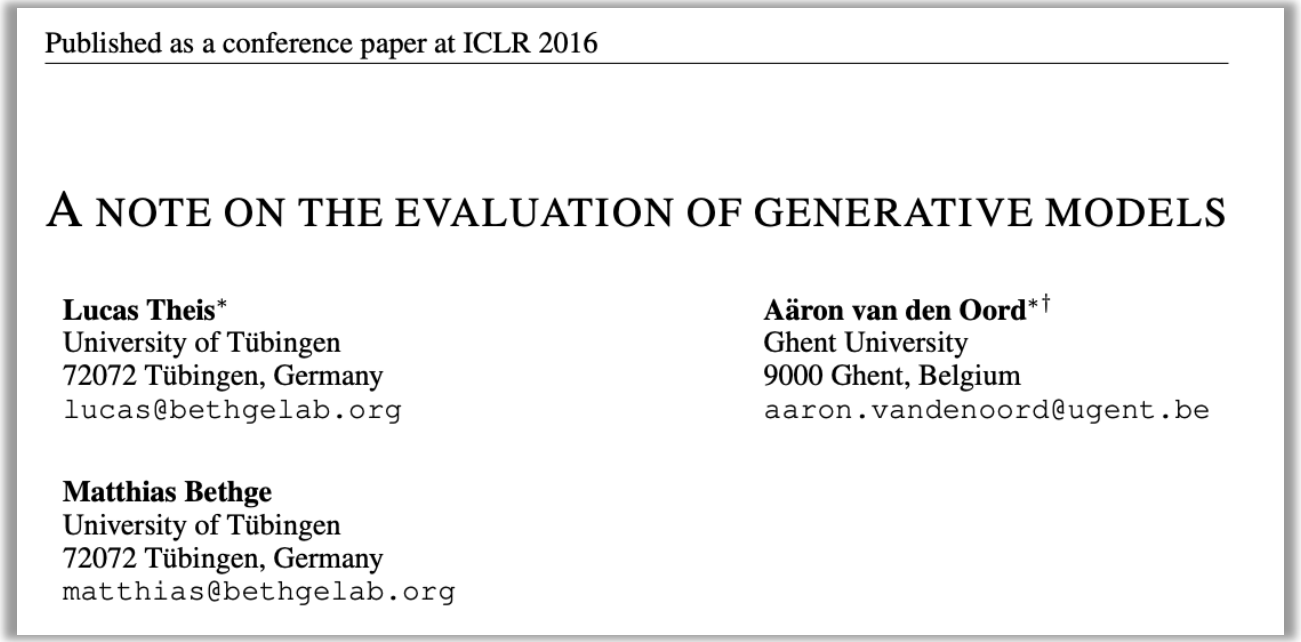
- **Pros and Cons using Likelihood**
  - **For Optimal Model ($p_{\theta^*} = p_{data}$)**
    - ✓ Highest test-time likelihood ↔ Best sample quality
  - **For Imperfect / Practical Model ($p_{\theta^*} \neq p_{data}$)**
    - ✓ Likelihood / D_{KL} is only **one possible distance metric** for $d(p_\theta, p_{\text{data}})$



Lucas Theis, Aäron van den Oord, Matthias Bethge. A note on the evaluation of generative models. ICLR 2016.

Towards Likelihood-Free Training

- **Pros and Cons using Likelihood**



Published as a conference paper at ICLR 2016

A NOTE ON THE EVALUATION OF GENERATIVE MODELS

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- **For Imperfect Model ($p_{\theta^*} \neq p_{data}$)**
  - ✓ Achieving high log-likelihoods might **not always imply** good sample quality, and vice-versa

Lucas Theis, Aäron van den Oord, Matthias Bethge. A note on the evaluation of generative models. ICLR 2016.

Towards Likelihood-Free Training

- **High Likelihood ≠ Good Sample Quality**
  - **Example 1: Great test log-likelihood, poor sample quality**

$$p_\theta(x) = 0.01\, p_{\text{data}}(x) + 0.99\, p_{\text{noise}}(x)$$

99% noise, bad quality

  - Taking logs gives the lower bound

$$\log p_\theta(x) \geq \log 0.01\, p_{\text{data}}(x) = \log p_{\text{data}}(x) - \log 100$$

  - Taking expectations over $x \sim p_{\text{data}}$, we have
    - ✓ **Lower bound:** $\mathbb{E}_{x\sim p_{\text{data}}}[\log p_\theta(x)] \geq \mathbb{E}_{x\sim p_{\text{data}}}[\log p_{\text{data}}(x)] - \log 100$
    - ✓ **Upper bound:** $\mathbb{E}_{x\sim p_{\text{data}}}[\log p_\theta(x)] \leq \mathbb{E}_{x\sim p_{\text{data}}}[\log p_{\text{data}}(x)]$

Towards Likelihood-Free Training

- **High Likelihood ≠ Good Sample Quality**
  - **Example 1: Great test log-likelihood, poor sample quality**

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$$\log p_\theta(x) \geq \log 0.01\, p_{\text{data}}(x) = \log p_{\text{data}}(x) - \log 100$$

  - Taking expectations over $x \sim p_{\text{data}}$, we have

$$\mathbb{E}_{x\sim p_{\text{data}}}[\log p_{\text{data}}(x)] - \log 100 \leq \mathbb{E}_{x\sim p_{\text{data}}}[\log p_\theta(x)] \leq \mathbb{E}_{x\sim p_{\text{data}}}[\log p_{\text{data}}(x)]$$

Towards Likelihood-Free Training

- **High Likelihood ≠ Good Sample Quality**
  - **Example 1: Great test log-likelihood, poor sample quality**

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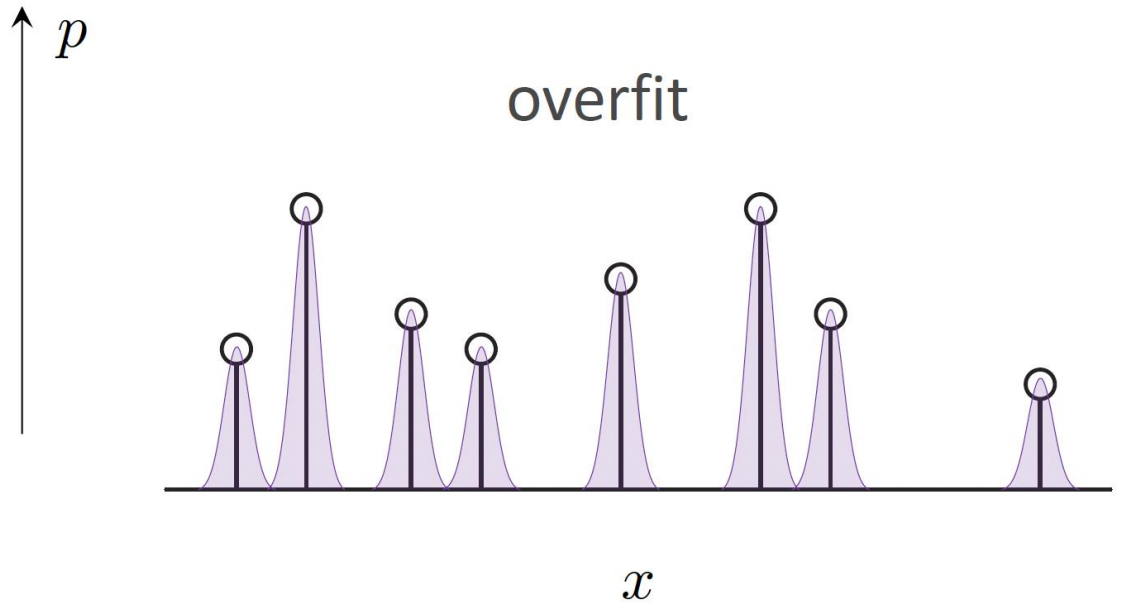
$$\mathbb{E}_{x\sim p_{\text{data}}}[\log p_{\text{data}}(x)] - \mathbf{\log 100} \leq \mathbb{E}_{x\sim p_{\text{data}}}[\log p_\theta(x)] \leq \mathbb{E}_{x\sim p_{\text{data}}}[\log p_{\text{data}}(x)]$$

  - **Question: What if x is of high-dimension (e.g, image)?**

$$\mathbb{E}_{x\sim p_{\text{data}}}[\log p_\theta(x)] \approx \mathbb{E}_{x\sim p_{\text{data}}}[\log p_{\text{data}}(x)]$$

Towards Likelihood-Free Training

- **High Likelihood ≠ Good Sample Quality**
  - **Example 2: Great sample quality, poor test log-likelihood**
  - Consider a model that simply memorizes the training set



Towards Likelihood-Free Training

- **High Likelihood ≠ Good Sample Quality**
  - **Example 2: Great sample quality, poor test log-likelihood**
  - Consider a model that simply memorizes the training set
    - ✓ **Generation quality**

      Sampling results have optimal quality ➔ Equals to real training data
    - ✓ **Test-time Likelihood**

      Test set will have zero probability assigned ➔ Worst test log-likelihood
  - Example 1 and 2 suggest **disentangling likelihoods and samples**

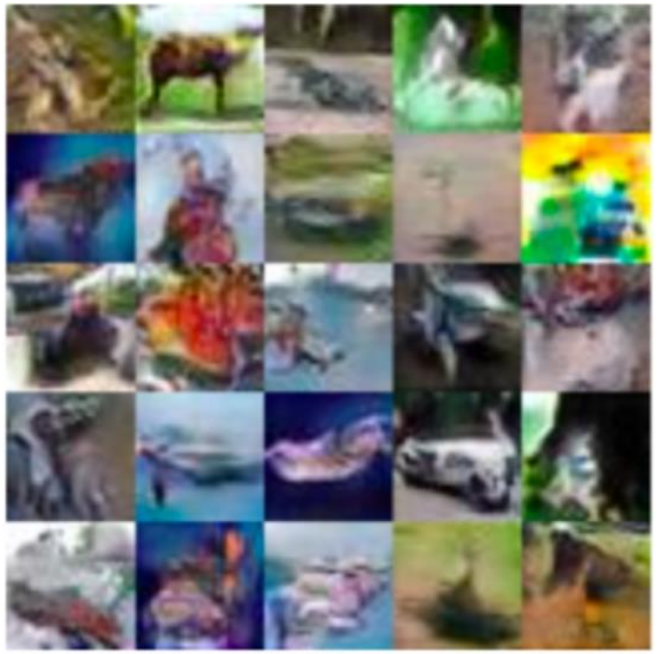
    ➔ **Likelihood-free learning**: objectives do not depend directly on likelihood

Towards Likelihood-Free Training

- **Comparing Distributions via Samples**
  - ➢ **Optimizing Likelihood ➔ Evaluating Generation Quality (directly)**



vs.



$S_1 = \{\mathbf{x} \sim P\}$ $S_2 = \{\mathbf{x} \sim Q\}$

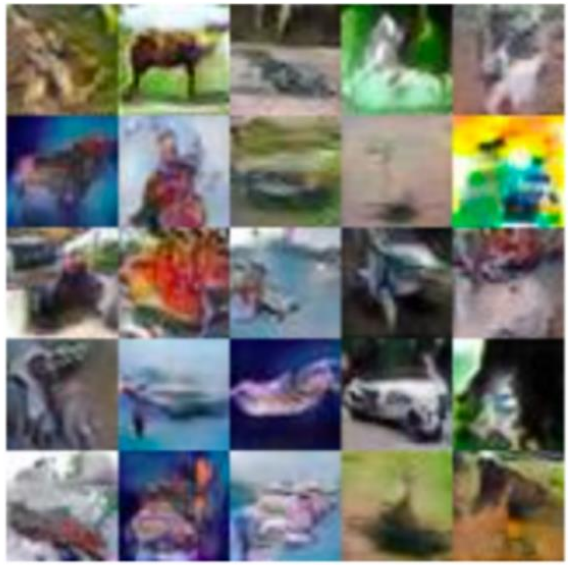
  - ➢ **Motivation:** Perform **sample-based evaluation** on generation quality

Towards Likelihood-Free Training

- **Comparing Distributions via Samples**
  - **Optimizing Likelihood ➔ Evaluating Generation Quality (directly)**



vs.



$S_1 = \{\mathbf{x} \sim P\}$ $S_2 = \{\mathbf{x} \sim Q\}$

  - **Question:** Given finite samples from two distributions $x \sim P$ and $x \sim Q$

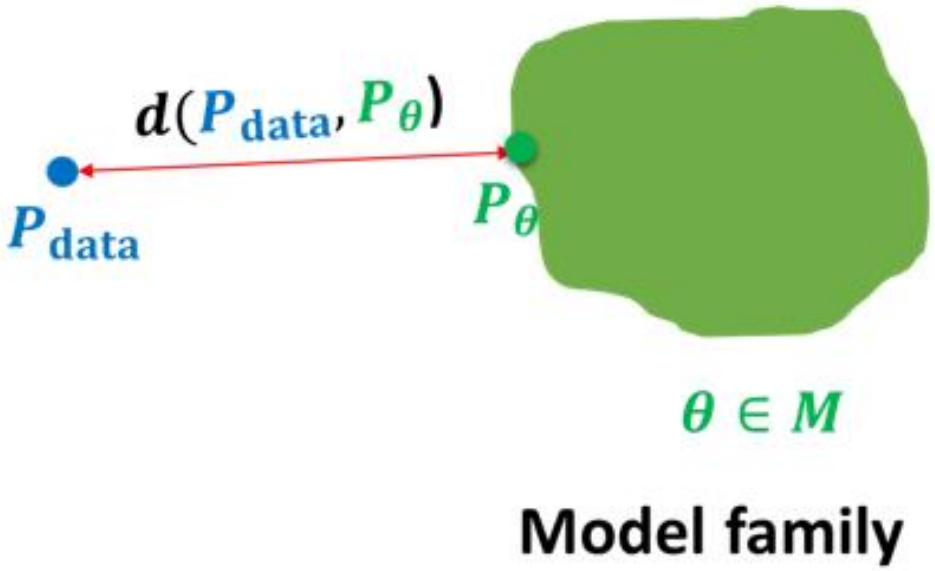
    ➔ Are these sample are from the same distribution ($P = Q$)?

Towards Likelihood-Free Training

- **Comparing Distributions via Samples**
  - ➢ **Optimizing Likelihood ➔ Evaluating Generation Quality (directly)**



$\mathbf{x}_i \sim P_{\text{data}}$
$i = 1, 2, \ldots, n$



  - ➢ **Question:** Given finite samples from two distributions $x \sim P_{\text{data}}$ and $x \sim P_\theta$

    ➔ Are these sample are from the same distribution ($P_{\text{data}} = P_\theta$)?

Towards Likelihood-Free Training

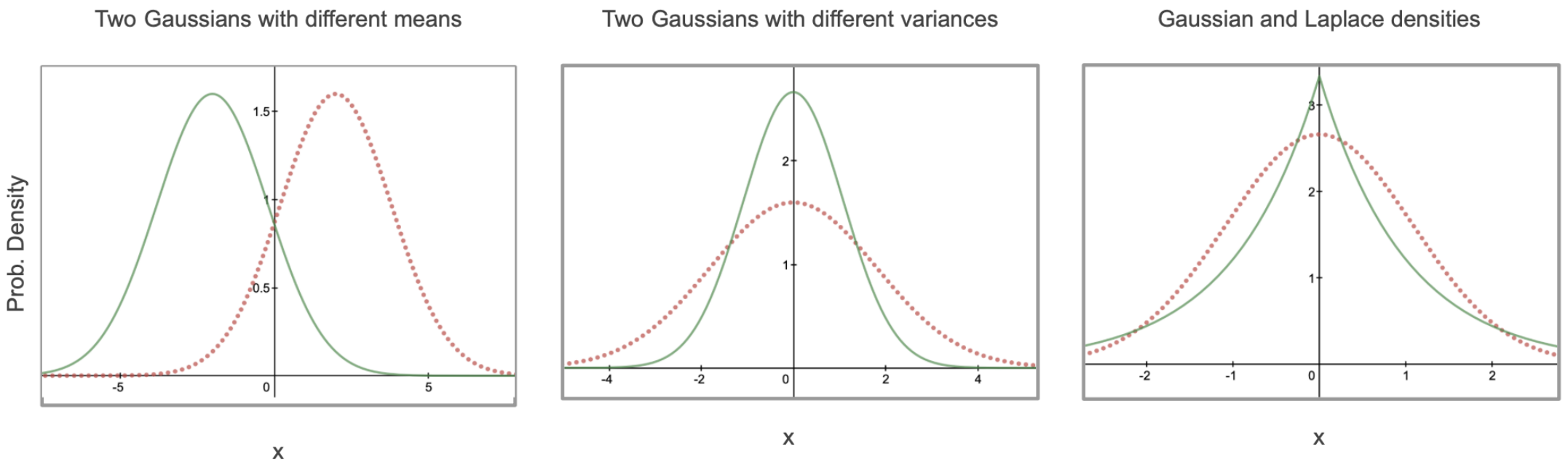
- **Two-Sample Tests (双样本检验)**
  - Given $S_1 = \{x \sim P_{\text{data}}\}$ and $S_2 = \{x \sim P_\theta\}$, a **two-sample test** consider
    - ✓ **Null Hypothesis (原假设):** $H_0: P_{\text{data}} = P_\theta$
    - ✓ **Alternative Hypothesis (备择假设):** $H_1: P_{\text{data}} \neq P_\theta$

Towards Likelihood-Free Training

- **Two-Sample Tests (双样本检验)**
  - Given $S_1 = \{x \sim P_{\text{data}}\}$ and $S_2 = \{x \sim P_\theta\}$, a **two-sample test** consider
    - ✓ **Null Hypothesis:** H_0: $P_{\text{data}} = P_\theta$
    - ✓ **Alternative Hypothesis:** H_1: $P_{\text{data}} \neq P_\theta$
  - **Test statistic T** compares S_1 and S_2 ➔ **Measurement of $d(P_{\text{data}}, P_\theta)$**
    - ✓ For example, difference in means or variance, etc.
    - ✓ If $T <$ threshold, then accept H_0 else reject it
  - **Key Observation:** Test statistic T is **likelihood-free** as no likelihood involved
  - **Goal:** Train P_θ to **minimize a two-sample test objective** between S_1 and S_2

Towards Likelihood-Free Training

- **Two-Sample Tests via a Discriminator (判別器)**
  - **Challenge:** A good test statistic T is hard to design



  - **Key idea:** **Learn a statistic** that maximizes a suitable notion of distance

    ➔ How? By **training a classifier** (i.e., **discriminator**) !



Lecture Outline

- **Towards Likelihood-Free Training**
- **Generative Adversarial Networks (GANs)**
- **Property of GANs**

Generative Adversarial Networks (GANs)

Generative Adversarial Nets

Ian J. Goodfellow,* Jean Pouget-Abadie,† Mehdi Mirza, Bing Xu, David Warde-Farley, Sherjil Ozair,‡ Aaron Courville, Yoshua Bengio§
Département d'informatique et de recherche opérationnelle
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Montréal, QC H3C 3J7

Abstract

We propose a new framework for estimating generative models via an adversarial process, in which we simultaneously train two models: a generative model G that captures the data distribution, and a discriminative model D that estimates the probability that a sample came from the training data rather than G. The training procedure for G is to maximize the probability of D making a mistake. This framework corresponds to a minimax two-player game. In the space of arbitrary functions G and D, a unique solution exists, with G recovering the training data distribution and D equal to $\frac{1}{2}$ everywhere. In the case where G and D are defined by multilayer perceptrons, the entire system can be trained with backpropagation. There is no need for any Markov chains or unrolled approximate inference networks during either training or generation of samples. Experiments demonstrate the potential of the framework through qualitative and quantitative evaluation of the generated samples.



Ian Goodfellow

Over 124k GS references

Generative adversarial networks

Authors: Ian Goodfellow, Jean Pouget-Abadie, Mehdi Mirza, Bing Xu, David Warde-Farley, Sherjil Ozair, Aaron Courville, Yoshua Bengio

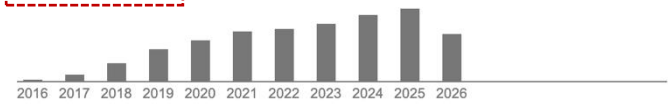
Publication date: 2014

Conference: Advances in neural information processing systems

Volume: 27

Description: We propose a new framework for estimating generative models via adversarial nets, in which we simultaneously train two models: a generative model G that captures the data distribution, and a discriminative model D that estimates the probability that a sample came from the training data rather than G. The training procedure for G is to maximize the probability of D making a mistake. This framework corresponds to a minimax two-player game. In the space of arbitrary functions G and D, a unique solution exists, with G recovering the training data distribution and D equal to 1/2 everywhere. In the case where G and D are defined by multilayer perceptrons, the entire system can be trained with backpropagation. There is no need for any Markov chains or unrolled approximate inference networks during either training or generation of samples. Experiments demonstrate the potential of the framework through qualitative and quantitatively evaluation of the generated samples.

Total citations: Cited by 124099



Scholar articles

Generative adversarial nets
IJ Goodfellow, J Pouget-Abadie, M Mirza, B Xu… - Advances in neural information processing systems, 2014
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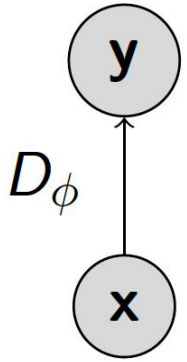
Generative adversarial networks *
I Goodfellow, J Pouget-Abadie, M Mirza, B Xu… - Communications of the ACM, 2020
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- "The most interesting idea in the last 10 years in machine learning"

— Yann Lecun, NIPS (2016)

Generative Adversarial Networks (GANs)

- **Two-Sample Test via Discriminator**



  - **Definition:** Any **binary classifier** $D_\phi(x) = P_\phi(y = 1 \mid x)$
    - ① "**real**" samples from the dataset $(y = 1)$
    - ② "**fake**" samples generated by the model $(y = 0)$
  - **Test statistic T: Negative Classifier Loss $-\mathcal{L}_{D_\phi}$**
    - ① **Low loss:** Real and fake samples are **easy to distinguish**
      - ➔ p_θ and p_{data} are different (value of T is high)
    - ② **High Loss:** Real and fake samples are **hard to distinguish**
      - ➔ p_θ and p_{data} are different (value of T is low)

  - **Goal: Minimizing $\mathcal{L}_{D_\phi}$**, or equivalently **maximizing T** (support $p_\theta \neq p_{data}$)

Generative Adversarial Networks (GANs)

- **Discriminator in GAN**

$$D_\phi(x) = P_\phi(y = 1 \mid x)$$

Probability of x is a real sample

  - **Training Objective**

$$\max_{D_\phi} V(p_\theta, D_\phi) = E_{\mathbf{x}\sim p_{\text{data}}}[\log D_\phi(\mathbf{x})] + E_{\mathbf{x}\sim p_\theta}[\log(1 - D_\phi(\mathbf{x}))]$$

$$\approx \sum_{\mathbf{x}\in S_1} \log D_\phi(\mathbf{x}) + \sum_{\mathbf{x}\in S_2} [\log(1 - D_\phi(\mathbf{x}))]$$

➔ **Binary classification** with the **cross entropy objective** (fixed p_θ)

  - **Optimal Discriminator**

$$D^*_\theta(\mathbf{x}) = \frac{p_{\text{data}}(\mathbf{x})}{p_{\text{data}}(\mathbf{x}) + p_\theta(\mathbf{x})}$$

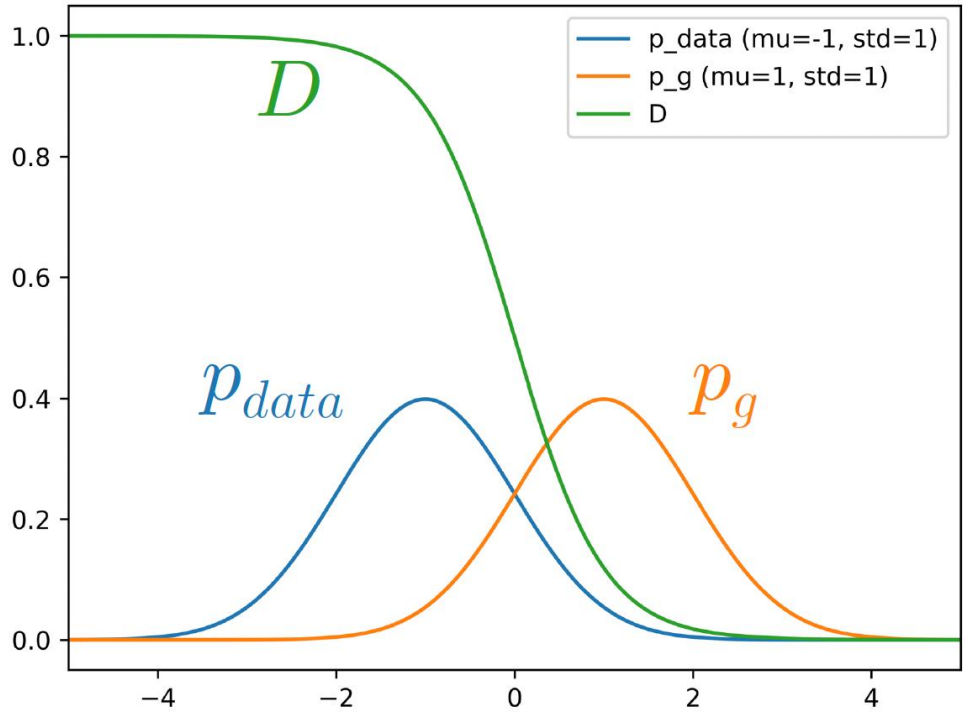
Maximize the cross entropy objective

✓ **Sanity Check:** If $p_\theta = p_{data}$, classifier cannot do better than chance (1/2)

Generative Adversarial Networks (GANs)

- **Discriminator in GAN**
  - **Optimal Discriminator**

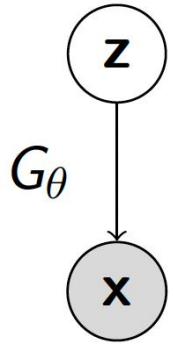
$$D_\theta^*(\mathbf{x}) = \frac{p_{\text{data}}(\mathbf{x})}{p_{\text{data}}(\mathbf{x}) + p_\theta(\mathbf{x})}$$



$$D_\phi^*(x) = P_\phi(y = 1 \mid x)$$

Generative Adversarial Networks (GANs)

- **Generator in GAN**
  - A **two-player minimax game** between **generator** and **discriminator**
  - **Generator:**

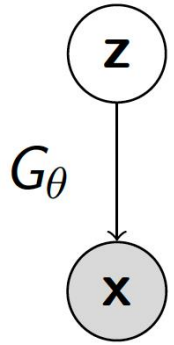


    - ✓ Directed, **latent variable model** with **deterministic** mapping $G_\theta : z \rightarrow x$

      ➔ Not required to be invertible!

Generative Adversarial Networks (GANs)

- **Generator in GAN**
  - A **two-player minimax game** between **generator** and **discriminator**
  - **Generator:**



    - ✓ Directed, **latent variable model** with **deterministic** mapping $G_\theta: z \rightarrow x$
    - ✓ Distribution $p_\theta(x)$ over x is **implicitly induced**

      ➔ Compared to VAE: $p_\theta(x) = \int p_\theta(x \mid z)\, p(z)\, dz$
    - ✓ Minimizes a two-sample test objective (supporting H_0: $P_\theta = P_{data}$)

Generative Adversarial Networks (GANs)

- **Generator in GAN**
  - **Training Objective**

$$\max_{D_\phi} V(p_\theta, D_\phi) = E_{\mathbf{x}\sim p_{\text{data}}}[\log D_\phi(\mathbf{x})] + E_{\mathbf{x}\sim p_\theta}[\log(1 - D_\phi(\mathbf{x}))]$$



$$\min_\theta \max_\phi V(G_\theta, D_\phi) = E_{\mathbf{x}\sim p_{\text{data}}}[\log D_\phi(\mathbf{x})] + E_{\mathbf{z}\sim p(\mathbf{z})}[\log(1 - D_\phi(G_\theta(\mathbf{z})))]$$

How to interpret this objective function?

Generative Adversarial Networks (GANs)

- **Generator in GAN**

$$\min_G \max_D V(G,D) = E_{\mathbf{x}\sim p_{\text{data}}}[\log D(\mathbf{x})] + E_{\mathbf{x}\sim p_G}[\log(1-D(\mathbf{x}))]$$

$$D^*_\theta(\mathbf{x}) = \frac{p_{\text{data}}(\mathbf{x})}{p_{\text{data}}(\mathbf{x}) + p_\theta(\mathbf{x})}$$

$$V(G, D^*_G(\mathbf{x})) = E_{\mathbf{x}\sim p_{\text{data}}}\left[\log \frac{p_{\text{data}}(\mathbf{x})}{p_{\text{data}}(\mathbf{x})+p_G(\mathbf{x})}\right] + E_{\mathbf{x}\sim p_G}\left[\log \frac{p_G(\mathbf{x})}{p_{\text{data}}(\mathbf{x})+p_G(\mathbf{x})}\right]$$

$$= E_{\mathbf{x}\sim p_{\text{data}}}\left[\log \frac{p_{\text{data}}(\mathbf{x})}{\frac{p_{\text{data}}(\mathbf{x})+p_G(\mathbf{x})}{2}}\right] + E_{\mathbf{x}\sim p_G}\left[\log \frac{p_G(\mathbf{x})}{\frac{p_{\text{data}}(\mathbf{x})+p_G(\mathbf{x})}{2}}\right] - \log 4$$

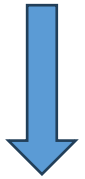
$$= \underbrace{D_{KL}\left[p_{\text{data}}, \frac{p_{\text{data}}+p_G}{2}\right] + D_{KL}\left[p_G, \frac{p_{\text{data}}+p_G}{2}\right]}_{2\times\text{Jensen-Shannon Divergence (JSD)}} - \log 4$$

$$= 2D_{JSD}[p_{\text{data}}, p_G] - \log 4$$

Generative Adversarial Networks (GANs)

- **Jenson-Shannon Divergence**
  - **JS Divergence:** Also called as the **symmetric KL Divergence**

$$\underbrace{D_{KL}\left[p_{\text{data}}, \frac{p_{\text{data}} + p_G}{2}\right] + D_{KL}\left[p_G, \frac{p_{\text{data}} + p_G}{2}\right]}_{2 \times \text{Jensen-Shannon Divergence (JSD)}}$$



$$D_{JSD}[p, q] = \frac{1}{2}\left(D_{KL}\left[p, \frac{p+q}{2}\right] + D_{KL}\left[q, \frac{p+q}{2}\right]\right)$$

"Total divergence to the average"

Generative Adversarial Networks (GANs)

- **Jenson-Shannon Divergence**
  - **JS Divergence:** Also called as the **symmetric KL Divergence**

$$D_{JSD}[p,q] = \frac{1}{2}\left(D_{KL}\left[p, \frac{p+q}{2}\right] + D_{KL}\left[q, \frac{p+q}{2}\right]\right)$$

  - **Properties**
    - ✓ **Non-negative:** $D_{JSD}[p,g] \geq 0$ (**Bounded:** $[0, \log 2]$)
    - ✓ **Symmetry:** $D_{JSD}[p,g] = D_{JSD}[g,p]$
    - ✓ $D_{JSD}[p,g] = 0 \text{ iff } p = q$
    - ✓ $\sqrt{D_{JSD}[p,g]}$ satisfies triangle inequality ➔ Jenson-Shannon Distance

Generative Adversarial Networks (GANs)

- **Jenson-Shannon Divergence**
  - **JS Divergence:** Also called as the **symmetric KL Divergence**

$$D_{JSD}[p, q] = \frac{1}{2}\left(D_{KL}\left[p, \frac{p+q}{2}\right] + D_{KL}\left[q, \frac{p+q}{2}\right]\right)$$

  - Objective of GAN

$$V\left(G, D_G^*(x)\right) = 2 \cdot D_{JSD}[p_{data}, p_G] - \log 4$$

    - ✓ Optimal generator for JSD

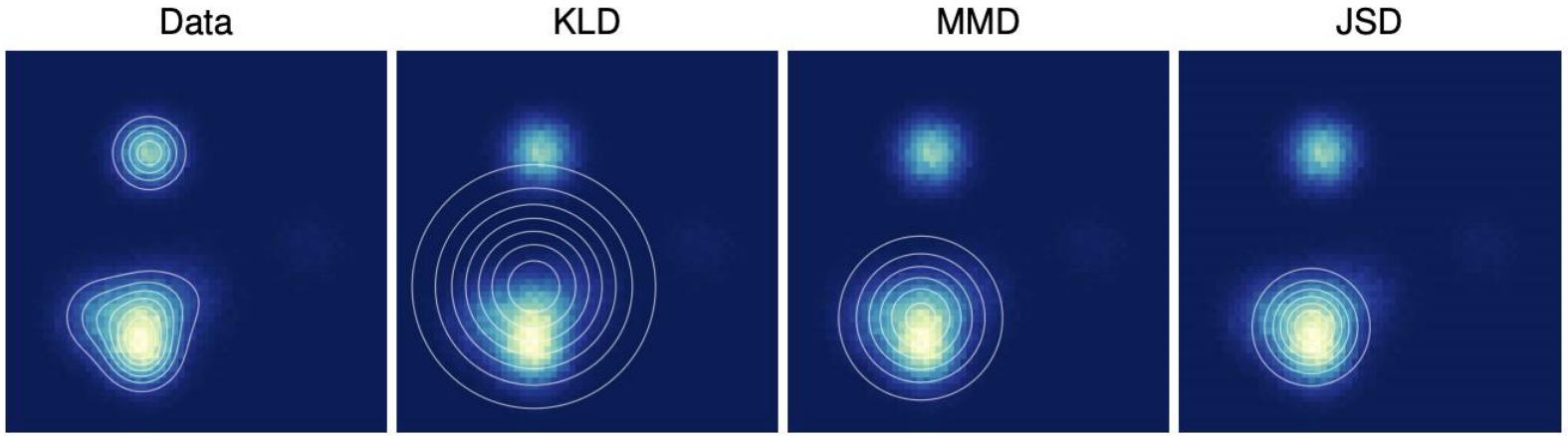
$$p_G = p_{data}$$

    - ✓ For the optimal generator and discriminator for JSD, we have

$$V\left(G, D_G^*(x)\right) = -\log 4$$

Generative Adversarial Networks (GANs)

- **Jenson-Shannon Divergence**
  - With imperfect p_θ, JS Divergence tends to perform **mode-seeking**



  - KL Divergence (log-likelihood objective) tries to average both modes

$$\mathbf{D}(P_{\text{data}}||P_\theta) = \mathbf{E}_{\mathbf{x}\sim P_{\text{data}}}\left[\log\left(\frac{P_{\text{data}}(\mathbf{x})}{P_\theta(\mathbf{x})}\right)\right] = \sum_{\mathbf{x}} P_{\text{data}}(\mathbf{x}) \log \frac{P_{\text{data}}(\mathbf{x})}{P_\theta(\mathbf{x})}$$

  - **JSD favors fitting one mode well (limited penalty for missing mode)**

Generative Adversarial Networks (GANs)

- **The GAN Training Algorithm**
  - Sample mini-batch of m training points $x^{(1)}, x^{(2)}, \ldots, x^{(m)}$ from $\mathcal{D}$
  - Sample mini-batch of m noise vectors $z^{(1)}, z^{(2)}, \ldots, z^{(m)}$ from p_z
  - **D-step:** Fix G_θ, optimize D_ϕ (stochastic gradient **ascent**)

$$\nabla_\phi V(G_\theta, D_\phi) = \frac{1}{m} \nabla_\phi \sum_{i=1}^{m} [\log D_\phi(\mathbf{x}^{(i)}) + \log(1 - D_\phi(G_\theta(\mathbf{z}^{(i)})))]$$

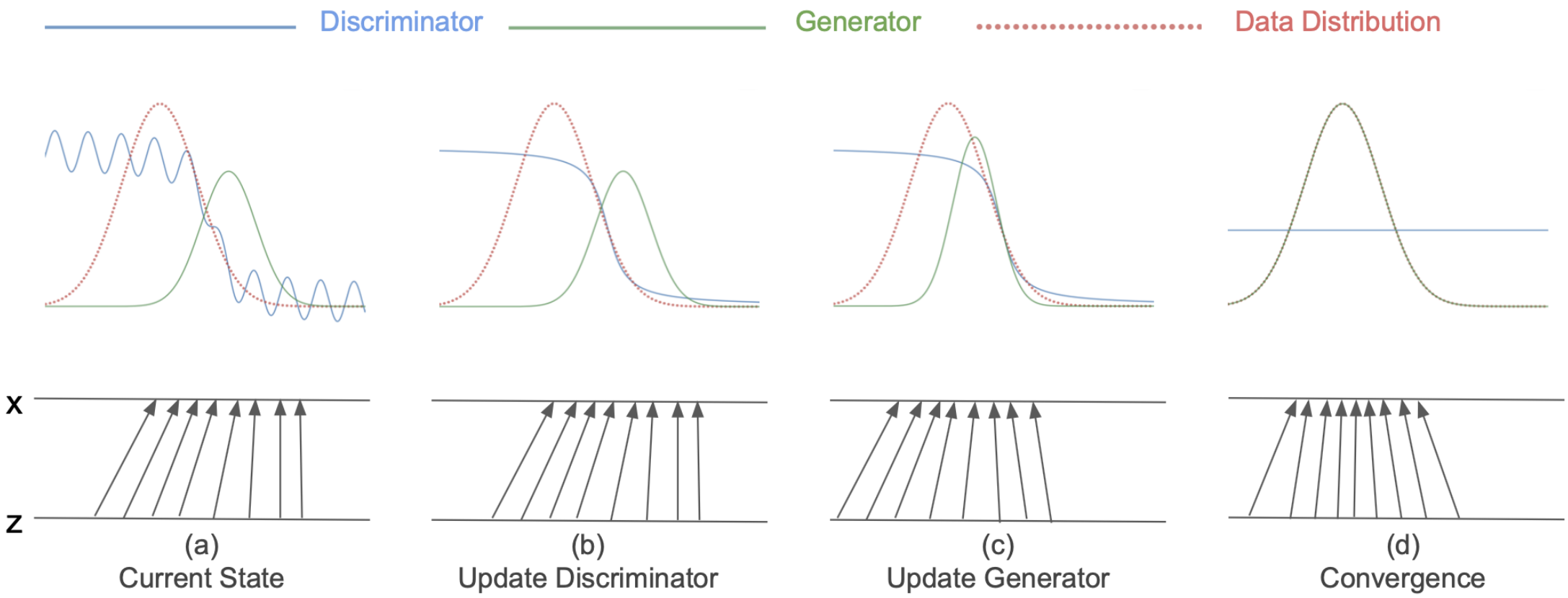
  - **G-step:** Fix D_ϕ, optimize G_θ (stochastic gradient **descent**)

$$\nabla_\theta V(G_\theta, D_\phi) = \frac{1}{m} \nabla_\theta \sum_{i=1}^{m} \log(1 - D_\phi(G_\theta(\mathbf{z}^{(i)})))$$

  - Repeat for a certain number of epochs (D-G Steps)

Generative Adversarial Networks (GANs)

- **The GAN Training Algorithm**
  - ➢ We can visualize training in 1D:



Content

- Towards Likelihood-Free Training
- Generative Adversarial Networks (GANs)
- Property of GANs

Property of GANs

- “Unprecedented” (史无前例的) High Generation Quality
  - ➢ Question: Which one is real?



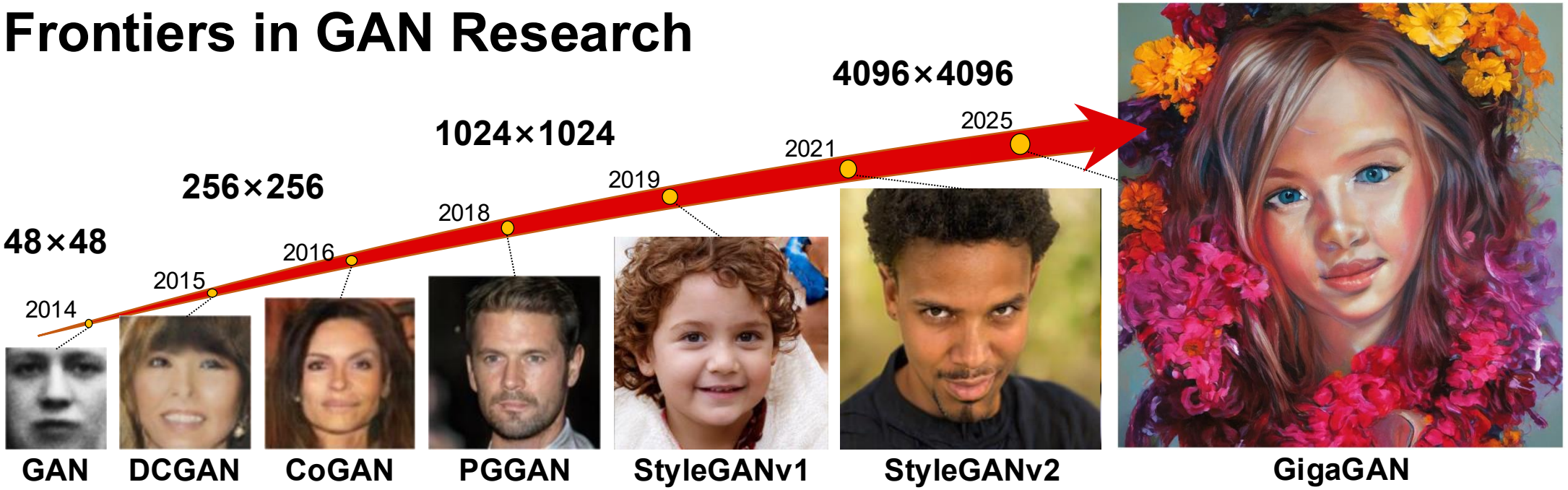
Source: Karras et al., 2018; The New York Times

Property of GANs

- **"Unprecedented" High Generation Quality**
  - **VAE vs. GAN**
    - ✓ **VAE:** Pixel-wise reconstruction ➔ Averaged / Blurred generation
    - ✓ **GAN:** Perceptual (感知的) similarity metric ➔ Realistic / Sharp results
  - **Readling List**
    - ✓ Larsen et al., "Autoencoding Beyond Pixels Using a Learned Similarity Metric," ICML 2016
    - ✓ Blau and Michaeli, "The Perception-Distortion Tradeoff," CVPR 2018
    - ✓ Theis et al., "A Note on the Evaluation of Generative Models," ICLR 2016

Property of GANs

- **Frontiers in GAN Research**

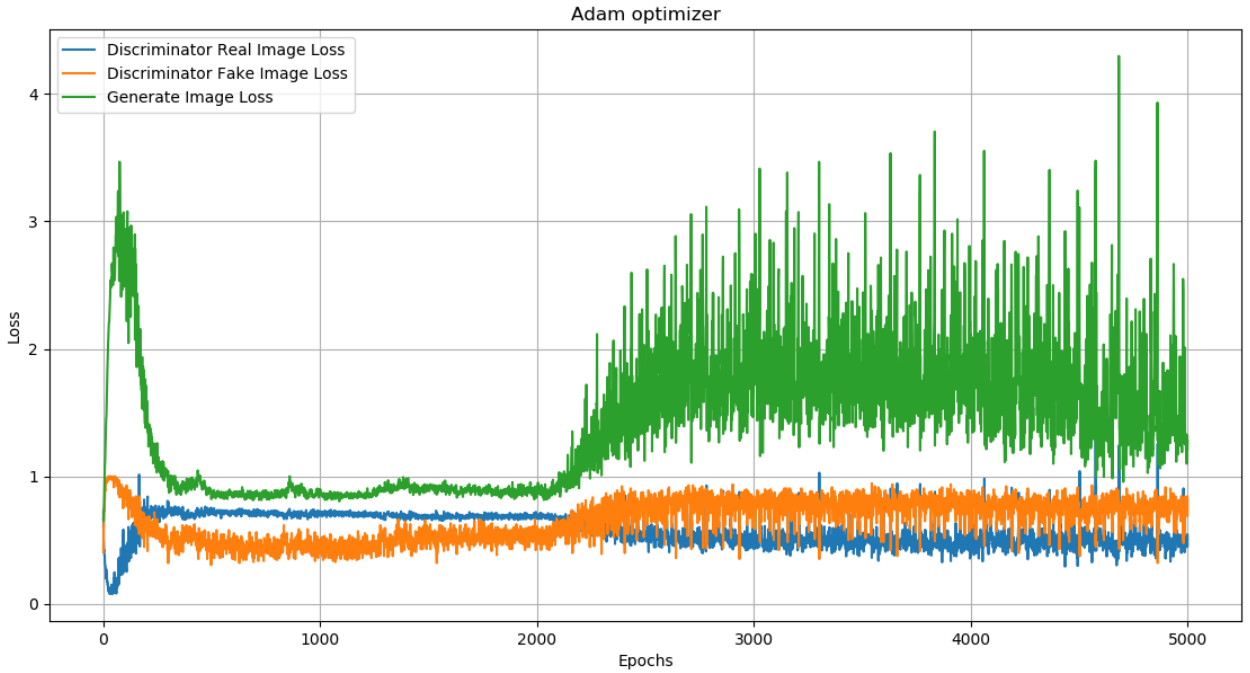


- ➢ GANs have been successfully applied to several domains and tasks
- ➢ However, working with GANs can be very challenging in practice
  - ✓ **Very unstable training / mode collapse / evaluation**
- ➢ Bag of tricks needed to train GANs successfully

Property of GANs

- **Optimization Challenges**
  - **Theorem:** G is guaranteed to converge to the data distribution, **IF** ① one can directly update p_G, and ② D is optimal at every step

    ➔**Unrealistic assumption!**



In practice:

- The loss value of G and D **keeps oscillating**
  - ✓ If it does not crash
  - ✓ Very easy to crash
- There is **no robust stopping criteria**

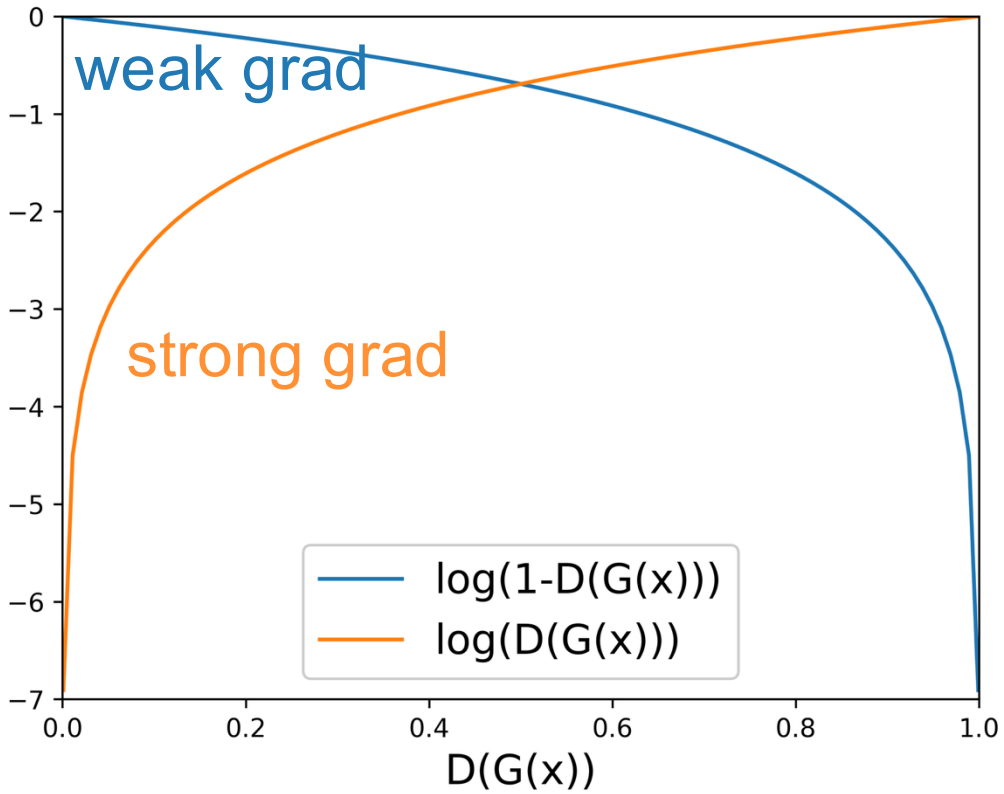
Goodfellow et al., "Generative Adversarial Nets," NeurIPS 2014, Sec. 4.2, Proposition 2.

Property of GANs

- **Generator in GAN**

$$\min_{\theta}\max_{\phi} V(G_\theta, D_\phi) = E_{\mathbf{x}\sim p_{\text{data}}}[\log D_\phi(\mathbf{x})] + E_{\mathbf{z}\sim p(\mathbf{z})}[\log(1 - D_\phi(G_\theta(\mathbf{z})))]$$

  - **The "Flip" Trick**



  - Early in training:
    - ✓ G is poor ➔ $D(G)$ is near 0
  - Thus, for gradients
    - ✓ $\nabla_G \log\big(1 - D(G)\big)$ **vanishes**
    - ✓ $\nabla_G - \log\big(D(G)\big)$ **persists**

    ➔ Maximizes the probability of being classified as "real" data

Property of GANs

- **Mode Collapse (模式坍塌)**
  - ➢ One of the most notorious limitation of GANs is "**mode collapse**"
  - ➢ Intuitively, mode collapse occurs when the generator produces samples from only a few modes of the data distribution, losing diversity

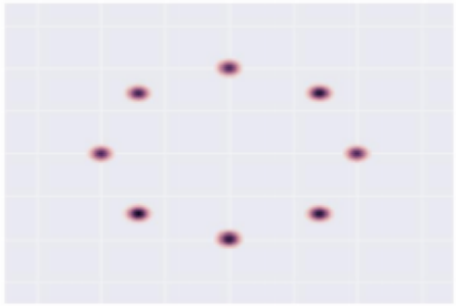


Arjovsky et al., 2017

Martin Arjovsky, Soumith Chintala, Léon Bottou. Wasserstein GAN. ICML 2017.

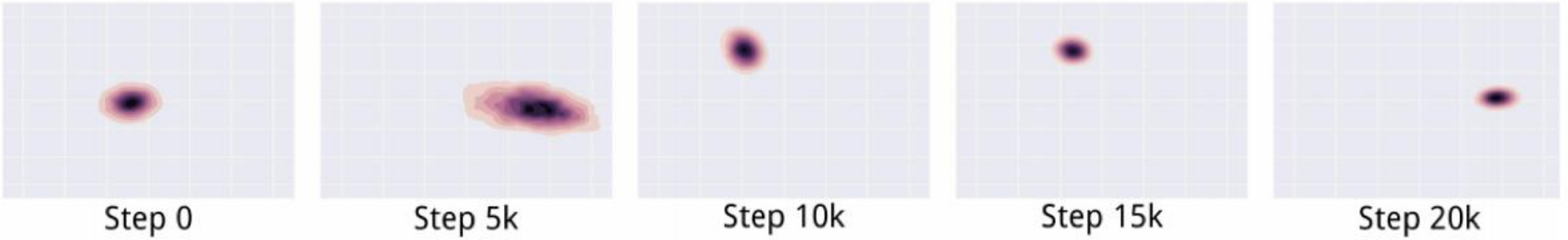
Property of GANs

- **Mode Collapse**
  - Suppose the true distribution is a mixture of Gaussians



Target

  - The generator distribution p_G keeps oscillating between different modes



Step 0 | Step 5k | Step 10k | Step 15k | Step 20k

Source: Metz et al., 2017

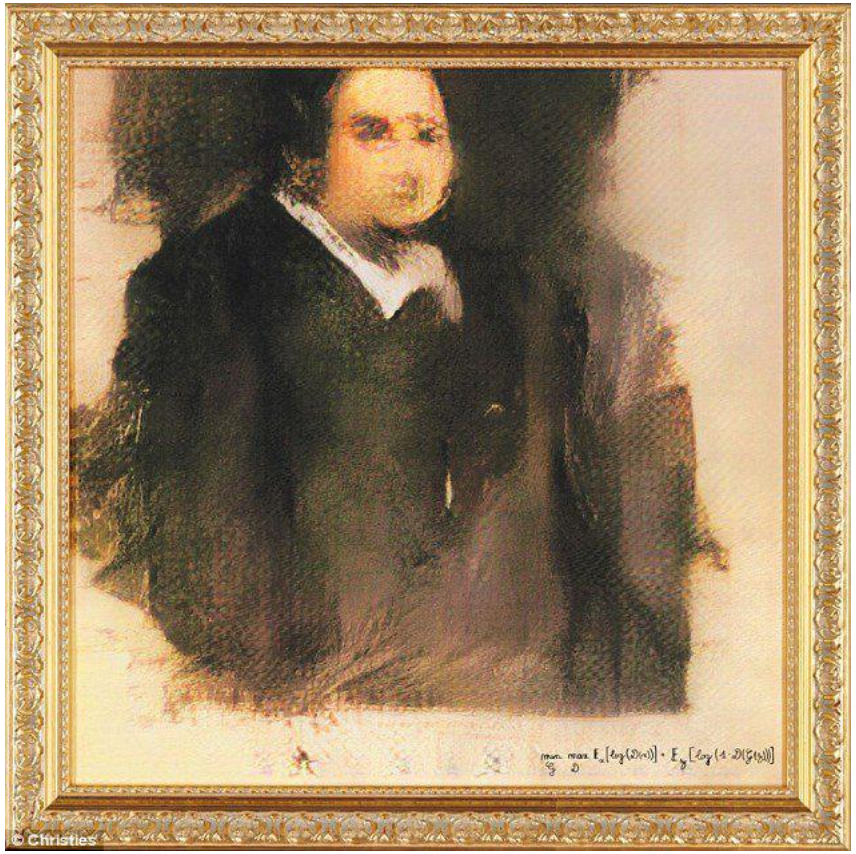
Luke Metz, Ben Poole, David Pfau, Jascha Sohl-Dickstein. Unrolled Generative Adversarial Networks. ICLR 2017

Property of GANs

- **Mode Collapse**
  - ➢ Fixes to mode collapse are mostly empirically driven
    - ✓ Alternate architectures
    - ✓ Regularization terms
    - ✓ Injecting small noise perturbations
    - ✓ …
  - ➢ **“ How to Train a GAN? Tips and tricks to make GANs work ”**
    - ✓ By Soumith Chintala
      - ➔ A co-founder of PyTorch
    - ✓ https://github.com/soumith/ganhacks

Property of GANs

- **“Beauty Lies in the Eyes of the Discriminator”**
  - ➢ Fixes to mode collapse are mostly empirically driven



“Portrait of Edmond de Belamy”

- ➢ Created by GAN
- ➢ Auctioned at Christie‘s in New York
  - ✓ **Expected Price:**

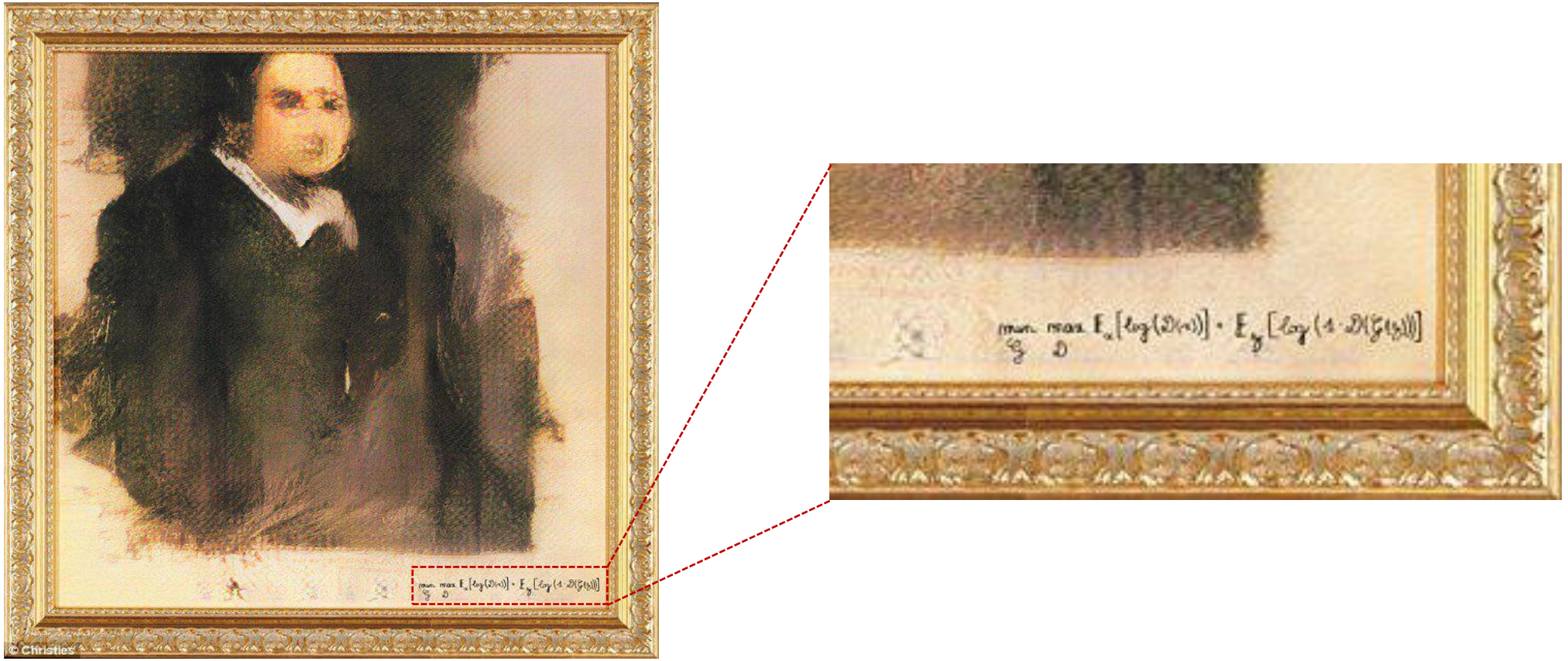
    $7,000 – $10,000
  - ✓ **True Price:**

    $432,500!

https://www.christies.com/about-us/features/A-collaboration-between-two-artists-one-human-one-a-machine-9332-7.aspx

Property of GANs

- **“Beauty Lies in the Eyes of the Discriminator”**
  - ➢ Fixes to mode collapse are mostly empirically driven



https://www.christies.com/about-us/features/A-collaboration-between-two-artists-one-human-one-a-machine-9332-7.aspx



Content

- **Towards Likelihood-Free Training**
- **Generative Adversarial Networks (GANs)**
- **Property of GANs**

Summary of Normalizing Flow Models

- **Key Motivation**
  - A two-player minimax game between the generator and discriminator
- **Pros**
  - Very realistic / high-quality / visually plausible generations
  - Very efficient at sampling
  - Very simple and intuitive framework, easy to adjust and improve
- **Cons**
  - Extremely difficult to train
  - Badly suffer from mode collapse

Supplementary Material

- **Stanford CS 236 Lecture 9**
  - https://www.youtube.com/watch?v=3Zv-gokhLu8&list=PLoROMvodv4rPOWA-omMM6STXaWW4FvJT8&index=9
- **Cornell CS 6785 Lecture 9**
  - https://www.youtube.com/watch?v=-3fzhDufRDc&list=PL2UML_KCiC0UPzjW9BjO-lW6dqliu9O4B&index=9
- **Deep Generative Models**
  - Chapter 8 Generative Adversarial Networks